

Kangwei Li

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About Me

I am currently pursuing a Bachelor of Science in Mathematics at Xi'an Jiaotong University in Xi'an, China. I intend to pursue a Ph.D. in Statistics or Operations Research upon completion of my undergraduate studies. My research interests include probability theory and duality in optimization.

Education

Xi'an Jiaotong University

Honors Science Program in Mathematics

Sept 2023 - Present

Xi'an, China

- GPA: 4.0/4.3
- Courses: *Real Analysis, Probability, Mathematical Statistics, Functional Analysis, Complex Analysis*

Harvard University

Visiting Undergraduate Student Program

Jan 2026 - Present

Cambridge, USA

- Courses: *Stochastic Process, Convex Optimization, Decision Theory, Probability II*

Hangzhou High School & Xi'an Jiaotong University

Honors Youth Program (Special Class for the Gifted Young)

Sept 2021 - Jun 2023

Hangzhou & Xi'an, China

Research Experience

Biologically plausible Neural Networks

Supervised by Prof. Jianyong Sun, Xi'an Jiaotong University

Jun 2025 - Sept 2025

Xi'an China

- Reviewed the shortcomings of backward algorithm
- Investigated biologically plausible neural network models, including equilibrium propagation
- Explored potential improvements to backpropagation from a biologically grounded perspective

Research on Wasserstein Policy Gradient Method

Advised by Prof. Rui Gao, University of Texas at Austin

Mar 2026 - Present

Remote

- Proposed Wasserstein-Euclidean flows for the Lagrangian of entropy-regularized constrained MDP.
- Explore the discretization of the flow to make the idea tractable.

Selected Awards & Honors

First Prize, The 16th Chinese Mathematics Competition(CMC)

Nov 2024

China Mathematical

Association

China National Scholarship (0.25% winning rate nationally)

Dec 2024

Xi'an Jiaotong University

Experience

AI for Mathematics Seminar

Peking University

Guangzhou, China

Jan 2025

- Learned Lean, a mathematical proof assistant and functional programming language.
- Finished a formalization of a theorem in convex optimization theory with Lean as a group project.

Skills

Programming Languages: Python, R, Lean

Language: English (Proficient), Mandarin (Native)